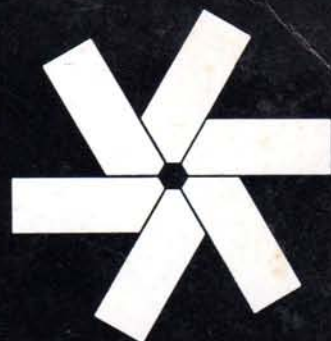


REMark[®]

January 1992

The Official Zenith Data Systems Computer Users' Magazine



The MastersPort 5Le



**Zenith Data Systems'
Newest and Most Innovative Laptop**

Perpetual Upgrading of a Z-248

Nick Visco
5639 Heming Avenue
Springfield, VA 22151-2707

I didn't start out to upgrade my venerable Z-248 to a 386SX, but as fortune would have it, I am certainly glad I did upgrade. Let's start at the beginning with my purchase of a Z-248 and go all the way to my present upgrade.

Before the Beginning

In February 1986, Zenith Data Systems was awarded a three year government contract to provide Z-248s to the government. In the spring of 1987, Zenith Data Systems decided to offer military and government workers the opportunity to buy Z-248s just above the price that they had been selling them to the government. The Z-248s were configured with dual floppy 5-1/4" disk drives, 512 KB of memory, EGA monitor, and optionally a 20 MB hard disk, just in case you ever thought you could fill one up. I decided that this was the perfect time to buy my first microcomputer. I will digress for a short while here for two purposes; first, to increase the length of this article to the required number of words, and secondly, to give you a little background of myself to let you identify with me in some small way.

In March of 1975, I started a masters program in something called computer science at the Naval Postgraduate School at Monterey, California. Gary Kildall, who wrote the 8-bit operating system CP/M, was one of my instructors, and Gordon Eubanks, who is president and CEO of Symantec, was a classmate. Microcomputers were just getting started, with 8-inch floppy disks from Shugart, Omron terminals with keyboard and monitor attached (4 KB of memory). I forget how much internal RAM was available. We had the familiar C> prompt, Debug.com, Edlin.com, Dir, and other ancient software (?) and

hardware. Altair computers were the rage of microcomputer buffs. I thought these little toys were cute, but the IBM 360 with punch cards (our military paychecks were printed on these every 15 days) and COBOL we had was real computing. I listened to Gary Kildall in class, and all about Intel 8008 and 8080 assembly code, which sounded very interesting, but FORTRAN was so easy (?) to use. In December of 1976, I completed a masters degree in computer science and went on with my job as a Naval Officer. I was sure Gordon and Gary would get real computer jobs with big computers in the future. I almost got it right, they got big computer jobs with real computers. Way to go guys!

The Beginning

I originally purchased my 8 MHz Z-248 in April of the 1987. The package included a 80286 processor, 512 KB of RAM, one 5-1/4" floppy (360 KB), one Seagate 20 MB (model 225) hard disk drive, one serial, and one parallel port, an EGA adapter (256 KB), a 84-key keyboard, ten 5-1/4" diskettes, MS-DOS 3.2, GW-BASIC, Windows 1.0, diagnostic software, and an EGA color monitor (Model ZVM-1380). I failed to mention the 10 slots (some already filled with a disk controller card, a CPU card, an I/O card, and a video card. This left 6 open slots). All this for the small price of \$2,482 which included tax, insurance, and delivery. I bought a copy of WordStar in June of 1987, a NEC P6 printer (which is still my only printer, no upgrades) in July of 1987, and I was on my way. In November of 1987, I received a surprise from Zenith Data Systems, a check in the amount of \$150. They had recently dropped the price of the CPU by \$100 and the monitor by \$50 and decided to give a

rebate to all those that had purchased their computers before the price decrease. I was happy and amazed that shortly over seven months after buying a product, that I considered worth every penny I spent, I would receive an unsolicited refund. I was hoping for more price cuts! No such luck.

As faith would have it, my Z-248 died in November of 1988. I opened up the machine and noticed that on power up, some of the green LEDs did not light up. It turns out that the I/O board was bad. Since the I/O has other functions besides just providing a serial and parallel port, I decided (I'm not sure if I had a choice) to purchase a replacement from Zenith Data Systems for a mere \$192. E.G come, E.G go.

Peripheral Upgrades

In July of 1988, I was assigned to the United States Naval Academy as an instructor in Computer Science. All instructors in the Computer Science department had a Z-248 on an Ethernet network with E-Mail on MILNET and BITNET to all colleges on the network. I checked out a 2400 bps Hayes modem from school. I thus had access to all the computers on campus that were on the network including my own if I left it in host mode. Plus, I had the ability to send E-Mail nationwide. These microcomputers had come a long way. I noticed in the computer trade journals that Gary and Gordon were doing quite well. Maybe they had been on to something big.

We taught Computer Literacy and PASCAL as an introduction course to all incoming plebes (freshmen). Since all plebes were required to purchase a Z-248 in August of 1986 and 1987, and a Z-286 in 1988 and 1989, I had zero compatibility problems during my two year tour. A LAN was installed in 1989 in Bancroft Hall where

all midshipmen live. This was great for instructors and maybe not so great for students. I could send assignments via E-Mail from my office or home to all my students using an automatically generated mail list, instead of individually. The students did get back at me by sending me E-Mail on weekends, asking many questions. Why were they studying on weekends anyway? I also used disk space on a local mini-computer to store sample programs that could be downloaded by the students and run on their computers. Completed assignments and homework were also required to be sent over the network. Programs were sent in source code only. I would compile and run the student written programs to prevent virus infection. One professor used to accept compiled programs until one of them infected his computer.

I quickly determined that my Z-248 would require moderate upgrading if I was to keep pace with the technology explosion going on all around me. I purchased a serial Logitech mouse and had to install a serial card since my original serial port was being used by the modem. I also purchased 128 KB of memory from First Capitol Computer to fill the memory from 512 KB to 640 KB. More about this memory upgrade later. I purchased a high density 3-1/2", 1.4 MB (not 1.44 MB as is commonly referred to; because 1 MB is 1 KB X 1 KB or 1,024 X 1,024 = 1,048,576 bytes = 1 MB. A high density disk has 2,880 sectors of 512 bytes which is 1,474,560 bytes. 1,474,560 divided by 1,048,576 is 1.40625 MB. If we try our math with two times 720 KB to get 1.44 MB we will also fail. Two times 720 KB (1,024 X 720) is 737,280 bytes and two times this number is 1,474,560 bytes which divided by 1,048,576 gives us the same answer of 1.40625 MB. You are probably asking why this is so important. Well it is not very important, unless you have just finished your article and are about a hundred words short) disk drive.

I purchased this 1.44 MB (oops! 1.4 MB) drive to be compatible with the midshipmen that I sponsored (i.e., gave them a place to crash on weekends) who would come over my house with these little disks and try to put them in my 5-1/4" 360 KB floppy drive. Unsuccessfully I might add. I also determined that with all the software that I was purchasing, the assignments I was generating, and all the lesson plans that I was creating, a 20 MB hard drive was not going to be enough. So I purchased a 40 MB Seagate model 251-1 hard drive. Zenith Data Systems, thank their technical expertise, not only provided a disk controller board that allowed two floppy drives, and two hard drives; but also allowed the space to fit each internal bay of the machine with required power wires from the 200 watt power supply. Of course I had to

purchase updated ROMs to access the 1.4 MB drive. No problem as I had already replaced numerous ROMs in the three labs that we had on campus for an identical upgrade.

Video Upgrade

Upon departing the Naval Academy and retiring from the Navy in June of 1990, I returned my Hayes modem and decided that I could not live without one, so I purchased a Practical Peripheral 2400 bps external modem. After retiring, I took a short vacation and tried to find out what I really want to do with my life. I decided I would become a computer consultant. The last time I checked, Gary and Gordon were doing quite nicely in this microcomputer business.

I set up shop in my basement with my trusty Z-248 and started to build up a clientele. I purchased Windows 3.0 for two purposes. One, to make it easier on my wife and daughter to point and select whatever program they desired, and two, for me to play solitaire. I mean for me to become familiar with Windows. Of course, I could only run Windows in real mode since I only had 640 KB of memory, but so what, I didn't have a need to run it in any other mode.

One day before a big assignment was due, my monitor died. No green power light on the front panel. I had seen (no pun intended) this problem before on some machines in the computer labs at the Academy. The power supply in the monitor was bad. I called some local repair places. They would fix it for \$150. What to do? Fix an EGA monitor that still has a good adapter card or go with the new technology, VGA? Well I decided to upgrade my computer. I purchased a 14" VGA monitor with .28 mm dot pitch and an inexpensive VGA card for \$329 and \$89 respectfully. Later I found a repair place that would have fixed my EGA monitor for \$89. But, I am glad I now have a VGA monitor. Smart of Zenith Data Systems to have a video card slot which can be replaced instead of a video port in the mother board. As with the floppy drive, hard disk drive, modem, memory, ROM, and mouse upgrade, I was back in action in no time. I was under a lot more time pressure this time, but no problems were encountered. My assignment was completed on time.

Memory Upgrade

I mentioned that, back in late 1988, I purchased a 128 KB memory card from First Capitol Computer in St. Peters, Missouri. I have had three big problems with this memory upgrade. The first problem I encountered was when I tried to install Prodigy on my computer. The install process would freeze up my computer. I tried various problem solving techniques (plain

vanilla AUTOEXEC.BAT and CONFIG.SYS files, booting from a floppy with the 360 KB as my A: drive and then the 1.4 MB drive as the A: drive, no luck). The solution that finally worked was to remove the 128 KB memory card and install Prodigy. Prodigy worked fine with just the original 512 KB of memory, but I had grown accustomed to having 640 KB of memory. I decided to reinstall the 128 KB memory card. To my surprise, Prodigy worked as well as all my software. When Prodigy updated their software a few months later, I ran into the same problem. This time I was ready. I took out the 128 KB memory card, installed Prodigy, and then reinstalled the 128 KB memory card. My next problem with the 128 KB memory card came when I tried to retrieve a WordPerfect document from either floppy disk drive. I could not retrieve the document without getting a file conversion error. I could save the document to my hard drive with no problem and retrieve it. I could save the document to my floppy disk but could not retrieve it using my machine. I could retrieve the document from the floppy disk if I placed it in another machine that had WordPerfect. I tried various problem techniques again with no success. When I removed my 128 KB memory card, this problem disappeared. I figured I could live with this known problem, so I reinstalled my 128 KB memory card. The third problem that I encountered with this card was when I decided to upgrade my operating system to MS-DOS 5.0. During the install process some of the files would not expand from their compressed form when being copied from the MS-DOS 5.0 floppy to my hard drive. I manually expanded the files that would not do so on installation. I guess these files were located in the region 512 KB to 640 KB since it happened to groups of files every so often. I did not attempt to reinstall MS-DOS 5.0 without my 128 KB memory card so this is only conjecture. But I still had a problem. When I tried to format a disk with the operating system on it, everything appeared to work. The problem was that I could not boot off this floppy. I removed the 128 KB memory card. I could now boot off of this floppy. I reinstalled my 128 KB memory card and decided to live with these three known problems. My guess is that the transient portion of COMMAND.COM that loads in 640 KB downward is somehow incompatible (timing?) with the 128 KB memory card. I called First Capitol Computer to see if this is a known problem with a known fix, but they never returned my call.

I decided now was the time to upgrade my memory beyond 640 KB and rid myself of this 128 KB memory card. I decided on an AST 286 memory card since they used standard SIMMs which I could use later in another machine. I purchased

Continued on Page 48

Box 51315, Indianapolis, IN 46251, tel. (800) 426-3475, or (317) 856-7571 in Indiana; and The Software Labs, 3767 Overland Avenue #112-115, Los Angeles, CA 90034, tel. (800) 359-9998. Catalogs are available from both vendors.

Some of the more popular online dialup services (in order of appearance above) are as follows. Check with the individual vendors for current pricing.

PRODIGY

445 Hamilton Avenue
White Plains, NY 10601
(800) 759-8000

CompuServe Information Service, Inc.
5000 Arlington Centre
Boulevard, Columbus, OH 43220
(800) 848-8990
(614) 457-8650 in Ohio

Genie

(GE Information Services, Dept. 02B)
401 North Washington
Street, Rockville, MD 20850
(800) 638-9636

BIX

(Byte Information Exchange)
One Phoenix Mill Lane
Peterborough, NH 03458-9990
(800) 227-1983

ZiffNet

One Park Avenue
New York, NY 10016
(800) 346-3247 ✱

Continued from Page 36

the card for \$99 with no memory. I called AST and asked if I could use 70ns 1 MB SIMMs instead of the 100ns chips recommended in the book. They refused to give me an answer. I decided to buy two 70ns 1 MB SIMMs anyway for \$40 each. I took out the 128 KB memory card, back-filled the 512 KB to 640 KB according to the instructions. My machine came up. I tried to enter Windows and got a memory parity error. I had to do a cold reboot, and when I tried to run mem/c to take a look at my memory configuration, I received another parity error check. I called AST. They told that the 70ns chips were the problem. I exchanged them for 100ns chips and had the same problem. I called AST. They now said that their board did not work with Z-248s (I had told the first technician that I had a Z-248) since the parity bit was opposite of what was on my original 512 KB of memory (bit 0 vice 7). I put back in my 128 KB memory card and decided I didn't need a memory upgrade after all.

CPU Upgrade

For my last upgrade, I purchased an

Classified

FOR SALE: COLLECTORS ITEMS: H-89 8-BIT COMPUTER W/DISK DRIVE AND H-77 DUAL DISK DRIVE. PERFECT CONDITION. ALL NOTEBOOKS, MANUAL, PROGRAM DISK, EXTRA DISK, ETC. CALL (312) 254-4029 OR WRITE W. DAVENPORT, 4516 ALBANY AVE., CHICAGO, IL 60632.

ZENITH DATA SYSTEMS CP/M-85 DESKTOP COMPUTER. KEYBOARD; MONITOR; SOFTWARE, GW-BASIC 2.0, MS-DOS VERS. 2, PASCAL COMPILER, MULTI-PAN, FORTRAN COMPILER. CALL (607) 589-4514. \$350 TAKES ALL.

SXZ-248 from Technology Specialists, Inc. (3110 Columbia Pike, Suite 303, Arlington, VA 22204, phone 703-521-1886 or 1-800-4-UPGRADE). This has turned my 8 MHz Z-248 with 640 KB (?) of memory into a 20 MHz 80386SX with 4 MB of memory, but that is another story. I finally was able to get rid of the 128 KB memory card. ✱

